

Day 3: Switch and Router Configuration Basics

Answers with Cisco Packet Tracer commands

Q1. Create VLANs 'Sales' (10) and 'Support' (20)

Add a 2960 switch in Packet Tracer, open its CLI and run:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# end
Switch# write memory
```

Q2. Assign ports to VLANs and verify

Fa0/1 - Fa0/3 go to Sales (VLAN 10) and Fa0/4 - Fa0/5 go to Support (VLAN 20). The **interface range** command configures several ports at once.

```
Switch# configure terminal
Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface range fastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
Switch# show vlan brief
```

Expected output of **show vlan brief**:

VLAN	Name	Status	Ports
1	default	active	Fa0/6 ... Fa0/24, Gig0/1, Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5

Q3. Connect two routers and configure a static route

Topology: PC-A (LAN1) - R1 ==== R2 - PC-B (LAN2). Use a crossover Ethernet cable (or serial DCE/DTE) between routers.

Device	Interface	IP Address	Subnet Mask
R1	Gi0/0 (LAN1)	192.168.1.1	255.255.255.0
R1	Gi0/1 (link)	10.0.0.1	255.255.255.252
R2	Gi0/1 (link)	10.0.0.2	255.255.255.252
R2	Gi0/0 (LAN2)	192.168.2.1	255.255.255.0

Router 1:

```
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# interface gigabitEthernet 0/1
```

```
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Router 2:

```
R2(config)# interface gigabitEthernet 0/1
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# interface gigabitEthernet 0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Verify: show ip route on R1 should show an 'S' entry for 192.168.2.0/24 via 10.0.0.2. A ping from PC-A to PC-B should succeed. (R2 needs the return route too, otherwise replies are dropped.)

Q4. Ping from Sales VLAN to Support VLAN

Why the ping fails:

- VLANs are separate Layer 2 broadcast domains. A switch never forwards frames between VLAN 10 and VLAN 20.
- The two VLANs use different IP subnets (e.g. 192.168.10.0/24 and 192.168.20.0/24). A host sees the destination is on a different network and sends the packet to its default gateway.
- A Layer 2 switch has no routing function and no gateway exists for the VLANs, so the packet is dropped. The result is 'Request timed out' or 'Destination host unreachable'.

Solution 1: Router-on-a-stick (one router link, trunk to the switch, sub-interfaces per VLAN)

```
! Switch: trunk port towards router
Switch(config)# interface gigabitEthernet 0/1
Switch(config-if)# switchport mode trunk

! Router: one sub-interface per VLAN
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# no shutdown
R1(config-if)# interface gigabitEthernet 0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# interface gigabitEthernet 0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Set the default gateway of Sales hosts to 192.168.10.1 and Support hosts to 192.168.20.1. The router now routes between the VLANs.

Solution 2: Layer 3 switch (SVIs, faster, no external router)

```
Switch(config)# ip routing
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
```

Comparison:

Feature	Router-on-a-stick	Layer 3 Switch
Routing done by	External router	Switch (SVI)

Speed	Limited by one link	Hardware speed
Cost / best for	Cheap, small networks	Larger campus networks